

AI for All

We Need to Talk, AI: a comic essay, carried into the lecture

Prof. Dr.-Ing. Mark Schutera | September 14, 2026 · 7 parts

Seven parts, one comic

Part	Section	The question it answers
I	Intro	why talk about AI at all
II	Basics	what AI is, and what made it work
III	Examples	where it already runs
IV	Chances	what it is good for
V	Risks	what it costs us
VI	Outlook	who gets to shape it
VII	Social Utopias	what we would want it for

cover, the section spiral

How to read this deck

weneedtotalk.ai



One panel caption, **one slide**. Six slides in a row share a heading and a picture: that is one page of the comic, and the six dots under the quote say which panel you are on. Every quote is lifted verbatim. The one italic line under the dots is mine, not theirs.

The comic is a free read at weneedtotalk.ai under CC BY-NC-SA 4.0. It is an essay, not a textbook: it argues. Argue back.

page 5, the authors at the table

[1]

1

Intro

pages 6 to 10

Knowing what future we would like to have

“

Knowing the future is impossible. But what we can do:
Knowing what future we would like to have. And then
work on it.

the opening spread

Our creatures have always turned against us

“

Our creatures have always turned against us. This applies to the Golem of Jewish mythology as well as to Frankenstein.



The fear is older than the technology. Hawking only gave it a deadline.

Golem, Frankenstein and the rubber boots

Our creatures have always turned against us

“

Neither magic nor natural sciences have ever helped to make artificial humans submissive and controllable.



The fear is older than the technology. Hawking only gave it a deadline.

Golem, Frankenstein and the rubber boots

Our creatures have always turned against us

“

Parents whose kids prefer to wear rubber boots instead of weather-proof sandals even when it's hot outside know that.



The fear is older than the technology. Hawking only gave it a deadline.

Golem, Frankenstein and the rubber boots

Our creatures have always turned against us

“

An AI Take-Over is a hypothetical scenario in which AI takes control of the planet – taking it away from us.



The fear is older than the technology. Hawking only gave it a deadline.

Golem, Frankenstein and the rubber boots

Our creatures have always turned against us

“

In 2011, Stephen Hawking said that ‘Success in creating AI would be the biggest event in history.’



The fear is older than the technology. Hawking only gave it a deadline.

Golem, Frankenstein and the rubber boots

Our creatures have always turned against us

“

‘Unfortunately, it might also be the last, unless we learn how to avoid the risks.’ How can we make AI systems both safe and beneficial?



The fear is older than the technology. Hawking only gave it a deadline.

Golem, Frankenstein and the rubber boots

Who is Julia

“

Good question. That's something I ask myself, now and then.



Data will not lie to you strategically. It will still lie to you.

Julia, the digit wall and the 404

Who is Julia

“

I love making sense of our world. I love to discover patterns in data, with code.



Data will not lie to you strategically. It will still lie to you.

Julia, the digit wall and the 404

Who is Julia

“

I love data's friendliness, their lack of strategic answering. If they are biased it's not their fault but ours.



Data will not lie to you strategically. It will still lie to you.

Julia, the digit wall and the 404

Who is Julia

“

Our brains are prone to some errors. Overconfidence.
And discrimination against people who seem to be
different from us.



Data will not lie to you strategically. It will still lie to you.

Julia, the digit wall and the 404

Who is Julia

“

Artificial Intelligence (AI) might have the potential to help us find answers to problems we are not able to solve.



Data will not lie to you strategically. It will still lie to you.

Julia, the digit wall and the 404

Who is Julia

“

For that reason, we need to demystify AI. Therefore, I'm writing this comic. Let your voice be heard.



Data will not lie to you strategically. It will still lie to you.

Julia, the digit wall and the 404

Who is Lena

“

Not one thing in particular, I guess.



Ask what the way we draw AI says about what we expect from it.

Lena, the iceberg of visual codes

Who is Lena

“

I love visual language. I love to encrypt thoughts into pictures by exploiting society's visual memory.



Ask what the way we draw AI says about what we expect from it.

Lena, the iceberg of visual codes

Who is Lena

“

Visuality is one of the most vivid encryption systems I know. We are constantly increasing and renewing it collectively.



Ask what the way we draw AI says about what we expect from it.

Lena, the iceberg of visual codes

Who is Lena

“

It's fun to mess around with visual codes by recomposing associations. Pictures start to shift meanings and interfere in society's processes.



Ask what the way we draw AI says about what we expect from it.

Lena, the iceberg of visual codes

Who is Lena

“

AI is a field that gives us an intimate view on society's desires and fears.



Ask what the way we draw AI says about what we expect from it.

Lena, the iceberg of visual codes

Who is Lena

“

Looking from different angles might help us to overcome commonly known perspectives. Let your view be seen!



Ask what the way we draw AI says about what we expect from it.

Lena, the iceberg of visual codes

Lessons from AI in Fiction

“



Every AI in the parade is a character. Not one of them is a system.

the five fictional robots, 1927 to 2014

Lessons from AI in Fiction

“



Every AI in the parade is a character. Not one of them is a system.

the five fictional robots, 1927 to 2014

Lessons from AI in Fiction

“



Every AI in the parade is a character. Not one of them is a system.

the five fictional robots, 1927 to 2014

Lessons from AI in Fiction

“



Every AI in the parade is a character. Not one of them is a system.

the five fictional robots, 1927 to 2014

Lessons from AI in Fiction

“



Every AI in the parade is a character. Not one of them is a system.

the five fictional robots, 1927 to 2014

Lessons from AI in Fiction

“

Seems, like we should not seek to replicate but to augment ourselves.



Every AI in the parade is a character. Not one of them is a system.

the five fictional robots, 1927 to 2014

2

Basics

pages 11 to 19

What is Artificial Intelligence?

“

Actually, the methods are not new. Just the techniques and the data.



The word intelligence is doing work here the technology has not earned.

the essence of a new situation

What is Artificial Intelligence?

“

At the moment, AI is mainly concerned with performing complicated but repetitive tasks.



The word intelligence is doing work here the technology has not earned.

the essence of a new situation

What is Artificial Intelligence?

“

If you define intelligence as the ability to capture the essence of a new situation...



The word intelligence is doing work here the technology has not earned.

the essence of a new situation

What is Artificial Intelligence?

“

... real AI needs to think more like us. With intentionality, with emotions, with conscience; capturing counterfactuals.



The word intelligence is doing work here the technology has not earned.

the essence of a new situation

What is Artificial Intelligence?

“

This being said, AI is still in its early stages (although it reads differently on the news).



The word intelligence is doing work here the technology has not earned.

the essence of a new situation

What is Artificial Intelligence?

“

Up to now, it's more adequate to speak of 'artificially augmented intelligence'.*



The word intelligence is doing work here the technology has not earned.

the essence of a new situation

Computer Power

“

What kind of techniques helped establish AI? One key indicator is computer power.



Moore said eighteen months. Training compute doubles in three and a half.

the cost and computer-power curve

Computer Power

“

Moore's law is the observation that the power of computers will double every 18 months.



Moore said eighteen months. Training compute doubles in three and a half.

the cost and computer-power curve

Computer Power

“

The amount of computer power used for training AI has gone up to 3,5 month-doubling time.



Moore said eighteen months. Training compute doubles in three and a half.

the cost and computer-power curve

Computer Power

“

The growth has largely been driven by computer hardware, optimized to train and run AI-models.



Moore said eighteen months. Training compute doubles in three and a half.

the cost and computer-power curve

Computer Power

“

This explains why AI has been growing faster and faster...



Moore said eighteen months. Training compute doubles in three and a half.

the cost and computer-power curve

Computer Power

“

... together with advances in data availability and algorithms (=the recipes) behind AI.



Moore said eighteen months. Training compute doubles in three and a half.

the cost and computer-power curve

Algorithms 1

“

A second key indicator are better algorithms. Now we can teach machines instead of programming them.



The shift is from writing the rules to supplying the examples.

a cat? a cat? a dog..?

Algorithms 1

“

Before, when intending to automate a task using AI, we had 1) to understand it completely and 2) to program it.



The shift is from writing the rules to supplying the examples.

a cat? a cat? a dog..?

Algorithms 1

“

The more complicated the task, the harder it was to automate.



The shift is from writing the rules to supplying the examples.

a cat? a cat? a dog..?

Algorithms 1

“

The breakthrough was in understanding how we learn to perform tasks.



The shift is from writing the rules to supplying the examples.

a cat? a cat? a dog..?

Algorithms 1

“

Nowadays, AI learns through trial and error, based on data we feed it. The more data, the more accurate it becomes.



The shift is from writing the rules to supplying the examples.

a cat? a cat? a dog..?

Algorithms 1

“

Modern AI applications achieve breathtaking results that we could hardly imagine a few years ago.



The shift is from writing the rules to supplying the examples.

a cat? a cat? a dog..?

Algorithms 2

“

What do these buzzwords even mean? Machine Learning; Deep Learning; Neural Networks (NN)...?



ML, DL and NN are nested, not three names for the same thing.

Algorithms 2

“

Machine Learning (ML) helps recognizing patterns based on existing data and algorithms: an important branch of AI.



ML, DL and NN are nested, not three names for the same thing.

Algorithms 2

“

(Note: The AI bookshelf includes other topics, as well.)



ML, DL and NN are nested, not three names for the same thing.

the AI bookshelf

Algorithms 2

“

Before Deep Learning (DL), we had to know a lot about the data we were programming. Otherwise, ML would break down.



ML, DL and NN are nested, not three names for the same thing.

Algorithms 2

“

DL uses artificial Neural Networks (NN) that learn patterns directly from the data they are fed.



ML, DL and NN are nested, not three names for the same thing.

Algorithms 2

“

Like our brain, NNs compare new information with objects they know. However, they have not yet been able to think of new objects.



ML, DL and NN are nested, not three names for the same thing.

Data 1

“

Data is the third element that makes AI so powerful.
But what exactly is data?



Anything you can record is data, which is why nothing is too trivial to collect.

the data word cloud

Data 1

“

Any form of raw fact or figure is data. Whether on paper or in electronic form.



Anything you can record is data, which is why nothing is too trivial to collect.

the data word cloud

Data 1

“

Data describes what we know and we can draw conclusions from it. Data can take the form of text, numbers, images, or sounds.



Anything you can record is data, which is why nothing is too trivial to collect.

the data word cloud

Data 1

“

The internet and mobile devices like smartphones, drones or simple sensors have made data abundant and far more valuable.



Anything you can record is data, which is why nothing is too trivial to collect.

the data word cloud

Data 1

“

The more data an algorithm is trained on, the better its results: conclusions, predictions, timing, actions.



Anything you can record is data, which is why nothing is too trivial to collect.

the data word cloud

Data 1

“

Better results = higher usability = more users = more power. Therefore ‘data is the new oil’.



Anything you can record is data, which is why nothing is too trivial to collect.

the data word cloud

Data 2

“

Okay, all AI applications need training data to get good results.



Four of the five steps are data work. Only the last one is the model.

the data pyramid, four steps of five

Data 2

“

Only with the invention of the internet, we were able to collect and store enough data to make AI work.



Four of the five steps are data work. Only the last one is the model.

the data pyramid, four steps of five

Data 2

“

Too much data for us to process. Just the right amount for AI.



Four of the five steps are data work. Only the last one is the model.

the data pyramid, four steps of five

Data 2

“

There is an adaption of Maslow's Pyramid of Needs.
Aimed at achieving AI solutions.



Four of the five steps are data work. Only the last one is the model.

the data pyramid, four steps of five

Data 2

“

Four out of five necessary steps have to do with data.



Four of the five steps are data work. Only the last one is the model.

the data pyramid, four steps of five

Data 2

“

AI is a journey that begins with data: collecting, storing, transforming and exploring it.



Four of the five steps are data work. Only the last one is the model.

the data pyramid, four steps of five

General AI

“

In the 1960s, AI pioneers hoped that machines could soon learn to think without human intervention.



Everything in the next four parts is narrow AI, not the thing the news means.

the two AI Winters

General AI

“

But from 1973 onwards, disappointment and criticism in the community, followed by pessimism in the press, led to the 1st ‘AI Winter’. A 2nd one followed.



Everything in the next four parts is narrow AI, not the thing the news means.

the two AI Winters

General AI

“

Until the 2000s, AI was a dirty word that ‘simply didn’t work’. But when data, hardware and infrastructure were ready in 2010, a wide range of AI applications followed.



Everything in the next four parts is narrow AI, not the thing the news means.

the two AI Winters

General AI

“

Despite their impressive progress and success, today's AI is narrow. Its tasks are often classification and need a lot of data and a lot of energy.



Everything in the next four parts is narrow AI, not the thing the news means.

the two AI Winters

General AI

“

No AI can represent causal relationships or integrate abstract knowledge, e.g., what objects are, what they are for, and how they are typically used.



Everything in the next four parts is narrow AI, not the thing the news means.

the two AI Winters

General AI

“

We really need to adjust our expectations and stop over-hyping AI. If we don't, we may find ourselves in another AI Winter.



Everything in the next four parts is narrow AI, not the thing the news means.

the two AI Winters

3

Examples

pages 20 to 27

Maps

“

Using location data from smartphones...



You supplied the training data this morning, on the way in.

two phones, the delta and the route

Maps

“

... the AI-application Maps can analyze how fast traffic is moving...



You supplied the training data this morning, on the way in.

two phones, the delta and the route

Maps

“

... in real-time.



You supplied the training data this morning, on the way in.

two phones, the delta and the route

Maps

“

Additionally, maps incorporate user-reported traffic incidents.



You supplied the training data this morning, on the way in.

two phones, the delta and the route

Maps

“

The proprietary algorithms of Maps have access to vast amounts of data.



You supplied the training data this morning, on the way in.

two phones, the delta and the route

Maps

“

This means, Maps can suggest the best and fastest routes for us.



You supplied the training data this morning, on the way in.

two phones, the delta and the route

Recommendation

“

All major internet companies – or ‘platforms’ – refine their products and processes using AI.



When the platform generates the product too, what is left to choose?

the component tag cloud

Recommendation

“

If we're looking for books or shoes, internet shops use AI so that our purchase decision gets easier.



When the platform generates the product too, what is left to choose?

the component tag cloud

Recommendation

“

Likewise, streaming providers use AI to analyze our consumer habits down to the last detail.



When the platform generates the product too, what is left to choose?

the component tag cloud

Recommendation

“

We benefit from improved suggestions and rediscoveries of old pearls which we have missed so far.



When the platform generates the product too, what is left to choose?

the component tag cloud

Recommendation

“

Also, the platforms use the knowledge of our habits to create and offer new products similar to previous successes...



When the platform generates the product too, what is left to choose?

the component tag cloud

Recommendation

“

...or to let AI autonomously stir those components together that have worked great elsewhere.



When the platform generates the product too, what is left to choose?

the component tag cloud

Social Networks

“

Those of us who use social networks use AI frequently,
without necessarily knowing it.



Invisible by design, which is the precondition for everything else here.

what a wonderful match

Social Networks

“

Social networks use AI to recommend friends, news, photos, tags, advertisements.



Invisible by design, which is the precondition for everything else here.

what a wonderful match

Social Networks

“

They even ‘curate’ the posts of your friends according to general or personal interests – with the help of AI.



Invisible by design, which is the precondition for everything else here.

what a wonderful match

Social Networks

“

Now imagine a network that knows even more than our habits, relations, thoughts, interests, likes and dislikes, our hobbies and whereabouts.



Invisible by design, which is the precondition for everything else here.

what a wonderful match

Social Networks

“

Our state of health or bank account for example.



Invisible by design, which is the precondition for everything else here.

what a wonderful match

Social Networks

“

In China, this already exists in the form of ‘WeChat’. Just recently, the Chinese government officially announced to supervise each group with 12+ members.



Invisible by design, which is the precondition for everything else here.

what a wonderful match

Self-driving Cars

“

Today, we can already activate the autopilot function which automatically and comfortably drives our car or flies our plane.



The blocker is not control. It is explanation.

the unforeseen circumstance

Self-driving Cars

“

But ultimately, we are responsible. The systems are not autonomous – or self-driving.



The blocker is not control. It is explanation.

the unforeseen circumstance

Self-driving Cars

“

Within a few years, this might change.



The blocker is not control. It is explanation.

the unforeseen circumstance

Self-driving Cars

“

We would have plenty of room to sleep, work, read or play, sober or drunk after a party.



The blocker is not control. It is explanation.

the unforeseen circumstance

Self-driving Cars

“

Or maybe not. Nearly every car accident involves some sort of unforeseen circumstances, the AI must confront for the first time.



The blocker is not control. It is explanation.

the unforeseen circumstance

Self-driving Cars

“

Moreover, user tests show that we need autonomous systems to explain their decision to feel safe. Not easy.



The blocker is not control. It is explanation.

the unforeseen circumstance

Identifying Images

“

A lot of AI work deals with identifying patterns. At an enormous scale.

● ● ● ● ● ●

If one system reads a tumour and a target, who aims it?

rifle 2017 next to Picasso 1910

Identifying Images

“

In 2015, computers made fewer mistakes than we do sorting images into predefined categories.



If one system reads a tumour and a target, who aims it?

rifle 2017 next to Picasso 1910

Identifying Images

“

Since 2018, everybody can use a free tool helping AI beginners detect images.



If one system reads a tumour and a target, who aims it?

rifle 2017 next to Picasso 1910

Identifying Images

“

While early image identification could be lifesaving...



If one system reads a tumour and a target, who aims it?

rifle 2017 next to Picasso 1910

Identifying Images

“

...sometimes it is life-threatening. The military is a huge fan of it, too.



If one system reads a tumour and a target, who aims it?

rifle 2017 next to Picasso 1910

Identifying Images

“

Unfortunately, AI is susceptible to hallucinations: caused by biased data or by hackers who, for example, make AI mistake guns for apes.



If one system reads a tumour and a target, who aims it?

rifle 2017 next to Picasso 1910

Emotional AI

“

Only $\sim 10\%$ of the emotional meaning of a message is conveyed through words.



Words are a tenth of the message, so a text interface reads a tenth of you.

ten per cent of the message

Emotional AI

“

To really understand us, you need our facial expressions, our tone of voice, our gestures plus our words.



Words are a tenth of the message, so a text interface reads a tenth of you.

ten per cent of the message

Emotional AI

“

As our relationship with AI is becoming more and more intimate...



Words are a tenth of the message, so a text interface reads a tenth of you.

ten per cent of the message

Emotional AI

“

... we need our AI devices (like assistant systems, cars, or robots) to sense and adapt to what we mean and not only to our words.



Words are a tenth of the message, so a text interface reads a tenth of you.

ten per cent of the message

Emotional AI

“

Indeed, Emotional AI could help us with some highly valuable tasks. Help autistic children, for example, to learn emotions.



Words are a tenth of the message, so a text interface reads a tenth of you.

ten per cent of the message

Emotional AI

“

Nonetheless, our freedom of thought might be at stake in hostile scenarios.



Words are a tenth of the message, so a text interface reads a tenth of you.

ten per cent of the message

Exoskeletons

“

Exoskeletons are an external framework we can wear to augment our natural physical ability and reduce strain and weaknesses.



A body is a moving target, so the controller has to keep learning.

elderly man or firefighter

Exoskeletons

“

But every one of us is different.



A body is a moving target, so the controller has to keep learning.

elderly man or firefighter

Exoskeletons

“

We need assistance that is specifically fit for us. We need to tailor our devices at an individual level.



A body is a moving target, so the controller has to keep learning.

elderly man or firefighter

Exoskeletons

“

AI can use real-time measurements of our body signals, our breathing rate or hip extension for example.



A body is a moving target, so the controller has to keep learning.

elderly man or firefighter

Exoskeletons

“

This way, the AI can get our individual profile right. AI can make our exoskeletons fit our needs – elderly man or firefighter.



A body is a moving target, so the controller has to keep learning.

elderly man or firefighter

Exoskeletons

“

Our bodies are different and constantly changing.
Nowadays, the only feasible way of translating this to
robotics is through the help of AI.



A body is a moving target, so the controller has to keep learning.

elderly man or firefighter

4

Chances

pages 28 to 36

Dealing with Big Data

“

Our power to analyze the tons of data we are now able to store and to collect can be a bottleneck.



The bottleneck was never storage. It is us.

hundreds of us against the volume

Dealing with Big Data

“

AI helps us deal with large and complex datasets in ways we have never seen before.



The bottleneck was never storage. It is us.

hundreds of us against the volume

Dealing with Big Data

“

Even with hundreds of us analyzing patterns in big data, the sheer volume simply overwhelms our capacities.



The bottleneck was never storage. It is us.

hundreds of us against the volume

Dealing with Big Data

“

Even if we could analyze data, but it proves to be exhausting and tedious, we can use rule-based AI systems – expert systems – that do the job for us.



The bottleneck was never storage. It is us.

hundreds of us against the volume

Dealing with Big Data

“

Traditional algorithms are bound to follow the same logic over and over, without flexibility or learning new paths with new data.



The bottleneck was never storage. It is us.

hundreds of us against the volume

Dealing with Big Data

“

Now enter AI. Because AI systems get smarter as more data is given to them, they are well suited for patterns and anomalies over the time.



The bottleneck was never storage. It is us.

hundreds of us against the volume

Efficiency

“

In many situations, AI is simply faster and more accurate than we are. Let's look at an example*.



Ninety-five against eighty-five. What would you need to know to believe it?

twenty lawyers against one AI

Efficiency

“

A legal platform carried out a competition between 20 experienced lawyers against a trained AI.



Ninety-five against eighty-five. What would you need to know to believe it?

twenty lawyers against one AI

Efficiency

“

Competitors had to review 5 legal contracts in 4 hours and identify 30 legal issues.



Ninety-five against eighty-five. What would you need to know to believe it?

twenty lawyers against one AI

Efficiency

“

They were rated according to how accurately they identified each issue. Who won?



Ninety-five against eighty-five. What would you need to know to believe it?

twenty lawyers against one AI

Efficiency

“

Humans achieved, on average, an 85% accuracy rate, the AI 95%. Humans took 92 minutes to complete the task, the AI... 26 seconds.



Ninety-five against eighty-five. What would you need to know to believe it?

twenty lawyers against one AI

Efficiency

“

The human lawyers who competed against the AI in the experiment said, the tasks were very similar to what lawyers do every day.



Ninety-five against eighty-five. What would you need to know to believe it?

twenty lawyers against one AI

Cognitive Biases

“

Our brain, as a result of evolution, was not optimized for rational decision-making or perfect diagnoses.



A shared distortion does not average out. It compounds.

gender-bias guiding you through the world

Cognitive Biases

“

Instead, we strive for a competitive degree of ‘fitness’ in our specific environments; quite fast and often satisfying.



A shared distortion does not average out. It compounds.

gender-bias guiding you through the world

Cognitive Biases

“

Our brain power is biologically limited; therefore, we work with heuristics, trust our gut feeling or store & retrieve wrong evidence.



A shared distortion does not average out. It compounds.

gender-bias guiding you through the world

Cognitive Biases

“

We call this cognitive bias. There are many of them and we share them to some degree.



A shared distortion does not average out. It compounds.

gender-bias guiding you through the world

Cognitive Biases

“

And this is what makes our distortions have large-scale consequences. Were they random, they would cancel out.



A shared distortion does not average out. It compounds.

gender-bias guiding you through the world

Cognitive Biases

“

Used wisely, AI could indeed help us improve our diagnoses and decisions.



A shared distortion does not average out. It compounds.

gender-bias guiding you through the world

Creativity

“

AI can save time otherwise wasted on stupid, tedious work for creativity. But it can do even more.



The machine generates. Judging stays with us, and judging is the part we enjoy.

just a generic style transfer

Creativity

“

When an AI sorts our photos, it is easier to make a photo-calendar with our children (in 28 positions) for our parents as a present.



The machine generates. Judging stays with us, and judging is the part we enjoy.

just a generic style transfer

Creativity

“

When an AI puts together a playlist of up-and-coming producers of house music, it may inspire our next DJ-set.



The machine generates. Judging stays with us, and judging is the part we enjoy.

just a generic style transfer

Creativity

“

We can also get creative directly with the AI, compiling data and coding it, in a narrow area. But the AI has no judgement.



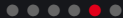
The machine generates. Judging stays with us, and judging is the part we enjoy.

just a generic style transfer

Creativity

“

Judging is an ability we possess. And we love doing it.
This is where AI together with us can achieve great
things, with a human tweak.



*The machine generates. Judging stays with us, and judging is the part we
enjoy.*

just a generic style transfer

Creativity

“

The smaller the role of humans in the process of art-creation becomes, the more AI questions traditional concepts of art.



The machine generates. Judging stays with us, and judging is the part we enjoy.

just a generic style transfer

Connection

“

Most doctors became doctors because they wanted to connect with people in order to heal them.



The chance is not the diagnosis. It is the time the diagnosis frees up.

the doctor as overpaid bookkeeper

Connection

“

Today, they often act like overpaid bookkeepers: taking in and spitting back data, prescribing drugs, adjusting doses, ordering tests.



The chance is not the diagnosis. It is the time the diagnosis frees up.

the doctor as overpaid bookkeeper

Connection

“

And it's still only retrospective, error-prone knowledge.



The chance is not the diagnosis. It is the time the diagnosis frees up.

the doctor as overpaid bookkeeper

Connection

“

But we cannot connect to doctors who struggle to keep up with messy data from disparate connections. And they cannot connect to us.



The chance is not the diagnosis. It is the time the diagnosis frees up.

the doctor as overpaid bookkeeper

Connection

“

AI can support proactive, individualized strategies.



The chance is not the diagnosis. It is the time the diagnosis frees up.

the doctor as overpaid bookkeeper

Connection

“

Our doctors could get to know us better, having the time to learn how a disease uniquely affects us and educate us in the right way.



The chance is not the diagnosis. It is the time the diagnosis frees up.

the doctor as overpaid bookkeeper

New Opportunities

“

AI allows for fascinating new opportunities. Teaching robots without a single line of coding, for example.



Show the robot instead of programming it: page 15's move, applied to a body.

the sensor jacket and glove

New Opportunities

“

Robots solve complex tasks today – but these tasks are deterministic. At any point of time, you know in advance where your robot needs to be.



Show the robot instead of programming it: page 15's move, applied to a body.

the sensor jacket and glove

New Opportunities

“

Every time a task or location changes, the robot's software must be re-programmed. This is time-consuming, expensive and difficult.



Show the robot instead of programming it: page 15's move, applied to a body.

the sensor jacket and glove

New Opportunities

“

Even for experienced programmers, it is hard to come up with a set of verbal rules of how to move when movements and settings are tricky or unpredictable.



Show the robot instead of programming it: page 15's move, applied to a body.

the sensor jacket and glove

New Opportunities

“

AI now enables us to teach robots how to move, either together with the robot, or, with jackets and gloves containing sensors.



Show the robot instead of programming it: page 15's move, applied to a body.

the sensor jacket and glove

New Opportunities

“

From the recorded example-movements, automation scripts can be created and optimized – without us having to write complex code. Fantastic!



Show the robot instead of programming it: page 15's move, applied to a body.

the sensor jacket and glove

Inclusion

“

AI is also about helping us, about improving inclusion and our well-being, by removing barriers.



Six barriers, one per panel, and not one of them is about convenience.

a wall missing several bricks

Inclusion

“

AI reads to us, it narrates the world around us and help us ‘see’. Speech-to-text AI helps us ‘hear’, too.



Six barriers, one per panel, and not one of them is about convenience.

a wall missing several bricks

Inclusion

“

AI helps us type (with words) or talks to us. Users of all ages are adopting virtual assistants because it is so simple to learn how to handle them.



Six barriers, one per panel, and not one of them is about convenience.

a wall missing several bricks

Inclusion

“

Smart home devices help us do things easily around the house and live independently.



Six barriers, one per panel, and not one of them is about convenience.

a wall missing several bricks

Inclusion

“

In the classroom, AI can identify individual challenges and offer more personalized approaches.



Six barriers, one per panel, and not one of them is about convenience.

a wall missing several bricks

Inclusion

“

AI virtual nurses or therapists offer 24-hour-support, or someone to talk to in case we are shy.



Six barriers, one per panel, and not one of them is about convenience.

a wall missing several bricks

Comfort

“

AI satisfies and reinforces our desire for comfort.



What did you stop leaving the house for, and what closed because of it?

mall_blueprint.pdf

Comfort

“

Almost all answers to our consumer, communication or information needs are just a mouse click or an app away. And most of the time, AI is involved.



What did you stop leaving the house for, and what closed because of it?

mall_blueprint.pdf

Comfort

“

It is tempting not to have to leave our couch at all, no matter what we want to do: eat, work, chat, watch a movie, listen to music...



What did you stop leaving the house for, and what closed because of it?

mall_blueprint.pdf

Comfort

“

'Pre-AI'-institutions such as shops, local transport, restaurants, libraries, or doctors must react to our new convenience.



What did you stop leaving the house for, and what closed because of it?

mall_blueprint.pdf

Comfort

“

All institutions with predominantly analogue offerings are undergoing a tremendous process of change. But it might be worth it.



What did you stop leaving the house for, and what closed because of it?

mall_blueprint.pdf

Comfort

“

It is precisely these institutions that are social places where we can come together and exchange ideas. Where we can decelerate. ‘Waste’ time.



What did you stop leaving the house for, and what closed because of it?

mall_blueprint.pdf

5

Risks

pages 37 to 43

Surveillance

“

AI makes monitoring people much easier. Not only via our words or facial expressions, via our consumer behavior or interests.



Not only where you are. What you are doing, through a wall.

the stick figures behind the wall

Surveillance

“

But also, via our movement patterns. Each one of us moves differently. Our movement patterns are unique.



Not only where you are. What you are doing, through a wall.

the stick figures behind the wall

Surveillance

“

Weak radio waves can penetrate walls, but our bodies reflect them. Using this data, an AI can be trained to identify our movements.



Not only where you are. What you are doing, through a wall.

the stick figures behind the wall

Surveillance

“

The AI can translate our movement patterns to little stick figures. Once registered, the AI can even identify us behind a wall.



Not only where you are. What you are doing, through a wall.

the stick figures behind the wall

Surveillance

“

That means the AI does not only know where we are.
The AI can also distinguish what we are doing, whether
we are standing, or eating.



Not only where you are. What you are doing, through a wall.

the stick figures behind the wall

Surveillance

“

It really comes close to magic. Hello, Harry Potter! And hello, surveillance state.



Not only where you are. What you are doing, through a wall.

the stick figures behind the wall

Opinion-Forming and Media

“

AI influences public opinion-forming and the media.
Our public sphere has become fragmented.



Who benefits from the current tools, and how would you find out?

the fragmented public sphere

Opinion-Forming and Media

“

Borders between private and public communication are blurry. User-generated content is rich but polarized and often privatized on closed platforms.



Who benefits from the current tools, and how would you find out?

the fragmented public sphere

Opinion-Forming and Media

“

New players and sources make special-interest-content popular and visible and, thus, important for mass media.



Who benefits from the current tools, and how would you find out?

the fragmented public sphere

Opinion-Forming and Media

“

Right-wing activists, for example, use AI bots, too, to spread hate and fake news in social media and, ultimately, mass media.



Who benefits from the current tools, and how would you find out?

the fragmented public sphere

Opinion-Forming and Media

“

Social media-users receive AI-personalized content of political organizations, based on users' personality profiles.



Who benefits from the current tools, and how would you find out?

the fragmented public sphere

Opinion-Forming and Media

“

Could we, in turn, use AI to enable a better opinion-forming – and identify those who benefit from current AI tools?*



Who benefits from the current tools, and how would you find out?

the fragmented public sphere

Future Work

“

The share of jobs at high risk of replacement by computers or robots has increased significantly.*



The list is not blue-collar. Lawyers, medics and brokers are on it.

the job ladder

Future Work

“

This involves all kinds of jobs on the social ladder.
White-collar jobs like lawyers, medics, or brokers...



The list is not blue-collar. Lawyers, medics and brokers are on it.

the job ladder

Future Work

“

... as well as creative jobs like artists, chefs or journalists, or blue-collar jobs such as construction workers or drivers.



The list is not blue-collar. Lawyers, medics and brokers are on it.

the job ladder

Future Work

“

AI can outperform us in doing repetitive tasks. Cheap, strong, fast, easy.



The list is not blue-collar. Lawyers, medics and brokers are on it.

the job ladder

Future Work

“

We are better in understanding non-trivial, new situations, building relationships, creating context and meaning. Custom-fit solutions.



The list is not blue-collar. Lawyers, medics and brokers are on it.

the job ladder

Future Work

“

We are also needed as partners of AI: developing algorithms that work, supervising, repairing. Combining us with AI.



The list is not blue-collar. Lawyers, medics and brokers are on it.

the job ladder

Data Security and Safety

“

Our blind trust in AI providers is misplaced. We all know that the amount of data that we are implicitly or explicitly forced to leave behind is too great.



Five of the six asks are legislation, not engineering.

the data letter inbox

Data Security and Safety

“

Since AI needs tons of data, we need to improve our data security and safety. A broad campaign, starting with, e.g.:



Five of the six asks are legislation, not engineering.

the data letter inbox

Data Security and Safety

“

The ‘data letter’, an obligation for all companies and authorities to tell us what data they store about us and how to view, correct and delete it;**



Five of the six asks are legislation, not engineering.

the data letter inbox

Data Security and Safety

“

Penalties for data outflows for system operators, and an obligation for connected devices to declare the frequency of the security updates (incentives);



Five of the six asks are legislation, not engineering.

the data letter inbox

Data Security and Safety

“

Permanent government investment in secure IT, e.g., open source components written and updated in secure programming languages;



Five of the six asks are legislation, not engineering.

the data letter inbox

Data Security and Safety

“

And: a widely available education in safe programming, along with educational, advisory and complaint services for the safe use of our digital systems.



Five of the six asks are legislation, not engineering.

the data letter inbox

Energy Use

“

In fact, AI has huge potential to optimize energy use, by analyzing complex data, making forecasts, optimizing processes via brain-like algorithms.



The comparison is with a brain, not with last year's data centre.

save energy, save the planet

Energy Use

“

But conventional hardware was not designed for AI, hence it needs much more computing power, much more energy, than our brains do.



The comparison is with a brain, not with last year's data centre.

save energy, save the planet

Energy Use

“

An AI supported translation service, e.g., DeepL, can handle 5,1 quadrillion operations per seconds. But that means masses of computing power and data.



The comparison is with a brain, not with last year's data centre.

save energy, save the planet

Energy Use

“

Today, data centers use more energy than some countries, e.g., Iran.* In 2018, we had 33 Zettabyte digital data per year. By 2025, we'll have 175.



The comparison is with a brain, not with last year's data centre.

save energy, save the planet

Energy Use

“

One of the most worrying forecasts says data centers will use 8% of global energy in 2030. The big internet firms bet on renewable energies.



The comparison is with a brain, not with last year's data centre.

save energy, save the planet

Energy Use

“

In short, our brains are far more energy efficient than our best computers – and we still do not really understand how.



The comparison is with a brain, not with last year's data centre.

save energy, save the planet

6

Outlook

pages 44 to 49

Overall Human-Machine Intelligence

“

We mainly interact with AI via our smartphone. A proprietary marketplace for countless uncured applications (=platform economy).



Your phone analyses you all day. What has it ever helped you think?

the smartphone as marketplace

Overall Human-Machine Intelligence

“

On the smartphone, we use AI basically for entertainment (image recognition), communication (language and translation) and information (e.g. google).



Your phone analyses you all day. What has it ever helped you think?

the smartphone as marketplace

Overall Human-Machine Intelligence

“

Unbeknownst to us, AI improves performance and battery life, too. And: It analyzes us, its users.



Your phone analyses you all day. What has it ever helped you think?

the smartphone as marketplace

Overall Human-Machine Intelligence

“

Up to date, there is no smartphone that is especially good at helping us get better ideas or interpret information. There is no interface for that either.



Your phone analyses you all day. What has it ever helped you think?

the smartphone as marketplace

Overall Human-Machine Intelligence

“

The development budgets are driven by numbers of purchase. In turn, our purchase decision is based on entertainment and communication values, and the design of the device.



Your phone analyses you all day. What has it ever helped you think?

the smartphone as marketplace

Overall Human-Machine Intelligence

“

But couldn't we also develop a smartphone that makes us more creative, more intelligent? That understands like us? Really augment us?



Your phone analyses you all day. What has it ever helped you think?

the smartphone as marketplace

Feminism

“

The contribution of women to AI is usually underestimated, especially concerning the democratic applicability of computing.



The first computers were women. The fear is not a takeover but a homogeneous room.

Ada Lovelace and Grace Hopper

Feminism

“

They were the first computers, actually: running calculations in the first information network of the world.



The first computers were women. The fear is not a takeover but a homogeneous room.

Ada Lovelace and Grace Hopper

Feminism

“

What we fear is not an AI-takeover. We fear scientists' homogeneity leading them to (finally!) 'create' intelligent beings – in an egocentric way.



The first computers were women. The fear is not a takeover but a homogeneous room.

Ada Lovelace and Grace Hopper

Feminism

“

In an AI world, where ‘female’ service robots are ‘hot’ and submissive, how can children learn to treat women with respect, with dignity?



The first computers were women. The fear is not a takeover but a homogeneous room.

Ada Lovelace and Grace Hopper

Feminism

“

And many of those whose service jobs will be replaced belong to an already vulnerable group: minority women.



The first computers were women. The fear is not a takeover but a homogeneous room.

Ada Lovelace and Grace Hopper

Feminism

“

We need (different!) women to participate in the development and use of AI. Otherwise we'll just reproduce patriarchal structures.



The first computers were women. The fear is not a takeover but a homogeneous room.

Ada Lovelace and Grace Hopper

Digital Capitalism

“

Algorithms have always been an integral part of our modern capitalist societies.



The paradigm is not production and sale. It is collection and use.

Weber's judge and Marx's worker

Digital Capitalism

“

The idea of objectiveness, general abstract rules, quantification, de-personalization and the concept of equality are characteristics of modernity.



The paradigm is not production and sale. It is collection and use.

Weber's judge and Marx's worker

Digital Capitalism

“

Max Weber, e.g., pictures the judge as an automation who processes concrete cases (data) based on an algorithm, namely the law.



The paradigm is not production and sale. It is collection and use.

Weber's judge and Marx's worker

Digital Capitalism

“

Marx' workers are an appendage of the machine, alienated from the spiritual potencies of the capitalist working process.



The paradigm is not production and sale. It is collection and use.

Weber's judge and Marx's worker

Digital Capitalism

“

Digital capitalism is the ground on which AI thrives. The central paradigm is – instead of production and sale – the collection and use of data.



The paradigm is not production and sale. It is collection and use.

Weber's judge and Marx's worker

Digital Capitalism

“

Since today's AI is largely incomprehensible to us, we need to regulate and shape AI as a public service and snatch the omnipotence over AI from the few dominant, digital companies.



The paradigm is not production and sale. It is collection and use.

Weber's judge and Marx's worker

ART: Accountability, Responsibility, Transparency

“

Throughout the world, we share a growing awareness:
We need to make AI safe, beneficial and fair for us.



The acronym is the easy part. The governance it asks for does not exist.

the three equal pillars

ART: Accountability, Responsibility, Transparency

“

This requires the participation and commitment from many of us. It means training, regulation and awareness in 3 equal pillars:



The acronym is the easy part. The governance it asks for does not exist.

the three equal pillars

ART: Accountability, Responsibility, Transparency

“

Accountability is the concept that AI should be held responsible for the results of its algorithms.



The acronym is the easy part. The governance it asks for does not exist.

the three equal pillars

ART: Accountability, Responsibility, Transparency

“

This includes our responsibility to develop a framework for AI that incorporates our values.



The acronym is the easy part. The governance it asks for does not exist.

the three equal pillars

ART: Accountability, Responsibility, Transparency

“

Transparency refers to the need to describe, inspect and reproduce the AI algorithms and results, and to manage the data used, in a fair way.



The acronym is the easy part. The governance it asks for does not exist.

the three equal pillars

ART: Accountability, Responsibility, Transparency

“

But let's not fool ourselves. For this ART of AI, we need a new and more ambitious form of governance.



The acronym is the easy part. The governance it asks for does not exist.

the three equal pillars

No Fear of Numbers

“

In order to understand how AI could improve our life,
we propose a real revolution:



Who in this room went quiet when the first equation appeared?

1908 against MCMVIII

No Fear of Numbers

“

Healing the public shaming in our individual math-biographies that caused fear.



Who in this room went quiet when the first equation appeared?

1908 against MCMVIII

No Fear of Numbers

“

Many of our brains signal ‘pain’ when we read numbers, think or talk about them.



Who in this room went quiet when the first equation appeared?

1908 against MCMVIII

No Fear of Numbers

“

Not only does this deprive life. Zero, e.g., is a fascinating tool allowing us to write ‘1908’ instead of MCMVIII: $1,000 + (1,000 - 100) + 5 + 1 + 1 + 1$.



Who in this room went quiet when the first equation appeared?

1908 against MCMVIII

No Fear of Numbers

“

Zero is also part of the binary code, fundamental to today's computers and their language. Fundamental to AI, therefore.



Who in this room went quiet when the first equation appeared?

1908 against MCMVIII

No Fear of Numbers

“

With improved ‘math-esteem’, more of us could and should discuss chances and risks of AI in our lives.



Who in this room went quiet when the first equation appeared?

1908 against MCMVIII

7

Social Utopias

pages 50 to 53

AI today is not Skynet

“

For over 300,000 years we've been hunting with spears;
for less than ten years we've been looking for our next
song with the help of AI.

the section opener

It depends on the training data

“

If we train AIs on discriminatory data, they will make discriminatory decisions and reduce, not increase, the opportunities, and resources of the underprivileged.

the application procedure

Raising the question, not settling it

“

An AI may not be able to resolve all injustices, but it can raise questions about how we want to live.

the motorway through the city

Or a creepy surveillance state

“

Or is this vision making us create a creepy surveillance state? Why should the authorities always be well-meaning? How could we make sure they are?

the same platform, read twice

What do you think?

“

But the first step has been made: What can AI help us with? Where do we have to be careful? What do you think? Let your voice be heard and your point of view be seen.

Tbc.

Further reading, the book's own

Authors	Title
Agrawal, Gans, Goldfarb	Prediction Machines: The Simple Economics of Artificial Intelligence, 2018
Evans	Broad Band: The Untold Story of the Women Who Made the Internet, 2018
Chollet, Allaire	Deep Learning with R, 2018
Lee	AI Superpowers: China, Silicon Valley, and the New World Order, 2018
Bostrom	Superintelligence: Paths, Dangers, Strategies, 2014
Zuboff	The Age of Surveillance Capitalism, 2018
Russell, Norvig	Artificial Intelligence: A Modern Approach, 2010
Daum	Das Kapital sind wir: Zur Kritik der digitalen Ökonomie, 2017

the reading list

[1]

If you want to build one

the track



This deck is about what AI does to us. For how it is built, work Google's **foundational courses** track, four courses in order:

Course	Length	Answers
Introduction to Machine Learning	20 min	what ML is
Machine Learning Crash Course	12 modules	how to build one
Problem Framing	45 min	whether to at all
Managing ML Projects	90 min	how to keep it alive

developers.google.com/machine-learning/foundational-courses

a laptop on the same table

[2]

References

- 1 Julia Schneider and Lena Kadriye Ziyal. *We Need to Talk, AI: A Comic Essay on Artificial Intelligence*. Berlin: Dr. Julia Schneider, 2019. ISBN: 978-3-748531-28-9. Free read, CC BY-NC-SA 4.0. <https://weneedtotalk.ai>.
- 2 Google Developers. *Foundational Courses: Machine Learning*. Course hub. <https://developers.google.com/machine-learning/foundational-courses>. 2025.
- 3 Google Developers. *Introduction to Machine Learning*. <https://developers.google.com/machine-learning/intro-to-ml>. 2025.
- 4 Google Developers. *Machine Learning Crash Course*. <https://developers.google.com/machine-learning/crash-course>. 2025.
- 5 Google Developers. *Introduction to Machine Learning Problem Framing*. <https://developers.google.com/machine-learning/problem-framing>. 2025.
- 6 Google Developers. *Managing Machine Learning Projects*. <https://developers.google.com/machine-learning/managing-ml-projects>. 2025.

**Let your voice be heard
and your point of view be seen**

Prof. Dr.-Ing. Mark Schutera | DHBW Ravensburg